



Biomechanical evaluation of human walking assisted by different knee exoskeleton configurations: a predictive simulation

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Abstract

Knee exoskeletons exhibit substantial variability in mechanical configurations and actuation pathways, leading to distinct biomechanical impacts on wearers. However, systematic cross-configuration comparisons under consistent modeling conditions remain limited. Using a unified predictive musculoskeletal framework, this study evaluates the human biomechanical responses to 3 representative knee exoskeleton configurations (*anterior cable-driven*, *lateral parallel-actuated*, *posterior linkage-driven*) during level-ground and uphill walking. Configuration-specific human–exoskeleton interaction forces were analytically modeled and applied to an identical musculoskeletal model under a standardized knee extension assistance profile. Across both terrains, all configurations reduced the knee extension moment impulse relative to unassisted walking, with the lateral configuration consistently achieving the largest reduction, reaching 33.6% (level ground) and 60.8% (uphill) compared to unassisted conditions. Quadriceps muscular effort exhibited pronounced terrain dependence, with the posterior configuration most effective on level ground and the lateral configuration during uphill walking. The posterior configuration consistently yielded the greatest reductions in peak knee contact force, with decreases of 6.3% (level-ground) and 16.5% (uphill) compared to unassisted conditions. All configurations reduced the cost of transport on both terrains, with the rankings across configurations closely following those of quadriceps muscular effort. These findings demonstrate that the effectiveness of knee exoskeleton assistance is governed not only by the assistance strategy, but also by the coherent interaction between force application geometry, joint posture, and task demands. The proposed predictive simulation framework provides a mechanical tool for identifying configuration-specific strengths and limitations to inform knee exoskeleton design prior to experimental deployment.

Keywords Human-exoskeleton interaction · Knee exoskeleton · Musculoskeletal simulation · Predictive simulation

1 Introduction

Wearable robots like knee exoskeletons in recent years have demonstrated the capability in facilitating human movement-related applications, making them a focal point of biomechanical and robotic research [1, 2]. With special focus on locomotion augmentation, researchers

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have developed a wide variety of knee exoskeletons in aid of knee flexion/extension, thereby achieving reductions in metabolic cost, improvements in gait efficiency, and decreases in the muscular effort required for level-ground or inclined locomotion [3, 4].

The complex anatomy and biomechanics of the knee joint, characterized by its multi-structured articulations, coupled rotational-translational movements, and the interplay of multiple muscle groups, have led to a diverse set of mechanical architectures of the knee exoskeleton [5, 6]. Existing knee exoskeletons exhibit substantial variability in their configurations, including actuation pathways and human-device interaction geometries, all of which lead to distinct biomechanical responses. For instance, cable-driven actuation designs, inspired by anisometric contractions of human muscles, typically route tensile forces across the knee, using pulleys or guides to generate assistive torques [2, 7, 8]; some knee exoskeletons are prototyped to deliver torques by a powered actuator located alongside the biological knee joint through a quasi-parallel kinematic chain [1, 9, 10]; other exoskeletons seek supportive assistance posterior to the knee, usually through linear actuators that extend during stance to assist knee extension [11, 12]. Beyond these, additional variations, such as continuum-structured designs [13], pneumatic pad-based devices [14], and semi-rigid chain-based configurations [15], have also been explored, further contributing to the diversity of knee exoskeleton configurations. Overall, these configurations differ systematically in actuation designs and force-distribution patterns, suggesting that their biomechanical consequences may vary substantially.

These distinctions highlight the importance of comparative evaluations across multiple knee exoskeleton configurations, particularly in understanding how different mechatronic architectures shape gait modulation, joint kinetics, and muscle recruitment strategies. Zantotto *et al.* investigated the dynamic consequences of varying thigh-link lengths in the ALEX II knee exoskeleton, revealing that link-length mismatches substantially affect the human-robot interaction torques and joint alignment during gait [16]. Other works have compared multiple knee exoskeleton configurations, ranging from pin-joint linkages to slider-based and cam-driven adaptive mechanisms, and demonstrated through dynamic modeling, simulation, and experimental verification that insufficient mechanical designs inevitably induce undesired internal joint loads [17, 18].

However, despite several studies examining the mechanical implications of individual configurations, existing analyses remain limited in both scope and depth. Cross-configuration investigations are still scarce, and the few available comparisons focus mainly on parallel-mechanism exoskeletons, with little attention devoted to other prevalent architectures such as cable-driven or posterior-actuated systems. Furthermore, most comparative studies do not extend their analyses to musculoskeletal responses or gait adaptations that ultimately determine the functional effectiveness and physiological impact of these devices [19]. Several practical and technical challenges hinder the development of comparative studies of this nature: firstly, experimentally prototyping and evaluating multiple exoskeletons demands considerable investments in device fabrication, safety verification, and other procedures that are prohibitively costly in both time and labor [20, 21]; in addition, variations in device architectures, including mechanical features, control strategies, and even strap positioning and tightness, make standardized, reproducible comparisons challenging to achieve across devices.

Predictive musculoskeletal simulation offers a compelling framework to overcome these challenges. Unlike traditional data-driven methods, this approach enables the investigation of human-exoskeleton interaction without the constraints of pre-recorded experimental kinematics. First, by integrating musculoskeletal models within a controlled virtual environment, researchers can systematically modulate assistive torques and interaction locations with high

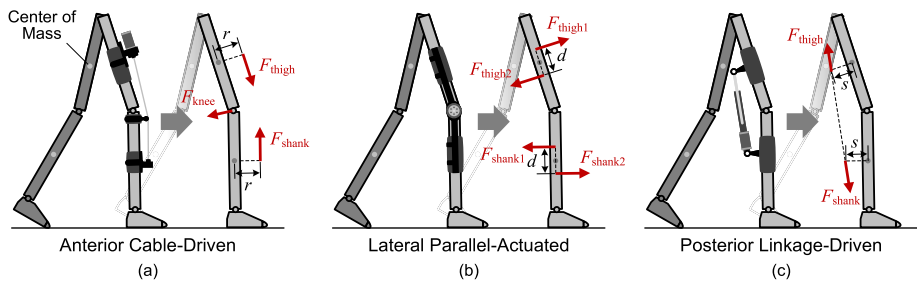


Fig. 1 Illustration of human-exoskeleton interaction for the three knee exoskeleton configurations studied. (a) *Anterior Cable-Driven*: A tensile cable routed anterior to the knee, produces a downward force on the thigh (F_{thigh}), an upward force on the shank (F_{shank}), and a posterior knee force due to cable redirection (F_{knee}). (b) *Lateral Parallel-Actuated*: Two lateral force couples, applied respectively on the thigh (F_{thigh1} , F_{thigh2}) and shank (F_{shank1} , F_{shank2}) with separation distance d . (c) *Posterior Linkage-Driven*: A posterior linkage transmits assistive torque via two posteriorly directed forces on the thigh (F_{thigh}) and shank (F_{shank}), separated by distance s

repeatability and minimal cost [22]. Second, since muscles serve as the primary actuators, this framework provides access to internal biomechanical quantities that are difficult to measure directly in vivo, such as individual muscle forces and metabolic cost, without requiring the additional distributional assumptions inherent in inverse-dynamics-based muscle force estimation methods [19]. Crucially, by employing muscle-reflex-based control strategies, these simulations can predictively generate gait adaptations induced by different exoskeleton configurations [23], thereby bypassing the need for experimental gait data as input [24].

This study extends cross-configuration investigations of knee exoskeletons beyond traditional kinematic or mechanical analysis to the musculoskeletal and gait-adaptation levels. We examine three representative exoskeleton configurations, characterize their distinct human-machine interaction forces under a unified assistive knee-extension torque, and apply these forces to a musculoskeletal model using a predictive simulation framework. This enables systematic evaluation of configuration-dependent effects on gait kinematics, joint kinetics, and muscular responses. To further examine the task-dependence of these effects, simulations are performed for both level-ground and uphill walking scenarios. Our work contributes to understanding how different exoskeleton configurations modulate human biomechanics and provides guidance for configuration selection and design in future exoskeleton development.

2 Methods

2.1 Human-exoskeleton interaction modeling

As shown in Fig. 1, we modeled the human-exoskeleton interaction forces for three representative designs: *anterior cable-driven* (AN) [2, 7, 8], *lateral parallel-actuated* (LA) [1, 9, 10], and *posterior linkage-driven* (PO) configurations [11, 12]. These configurations are selected because they capture the three dominant actuation pathways and force-distribution patterns prevalent in contemporary knee exoskeletons [25]. Each configuration is mechanically simplified to its essential force-transmission elements to facilitate consistent integration into the musculoskeletal simulation framework. For modeling tractability, the biological knee is represented as an ideal hinge with a fixed center of rotation, and exoskeleton mass, inertia, and transmission loss are neglected.

2.1.1 Anterior cable-driven configuration

In an AN system, a tensile force travels along the front of the thigh and shank, routed over a pulley near the knee joint (Fig. 1(a)). This routing generates three interaction forces on the wearer: a downward force on the thigh (F_{thigh}), an upward force on the shank (F_{shank}), and a posteriorly directed force at the knee due to cable redirection (F_{knee}). Assuming that the cable maintains an equal offset from the longitudinal axes of the thigh and shank, which equals to r , the forces F_{thigh} and F_{shank} act parallel to their respective limb segments. Because the cable transmits a uniform tensile load along its entire length, the magnitudes of the forces on the thigh and shank satisfy

$$F_{\text{thigh}} = F_{\text{shank}}. \quad (1)$$

Force equilibrium (i.e., the exoskeleton does not impose net linear momentum to the limb) further requires a posterior redirected force at the knee, given by

$$F_{\text{knee}} = 2F_{\text{thigh}} \sin(\theta_{\text{knee}}/2), \quad (2)$$

where θ_{knee} is the angle of the knee joint (positive in flexion). The net effect produces an assistive knee-extension torque

$$\tau_{\text{exo}} = F_{\text{thigh}} \cdot r. \quad (3)$$

2.1.2 Lateral parallel-actuated configuration

In an LA system, the actuator and linkage are mounted alongside the limb and transmit interaction forces via lateral braces (Fig. 1(b)). The exoskeleton applies assistive torque through two force couples: one acting on the thigh and one on the shank [10]. Each couple consists of two equal and opposite lateral forces acting normal to the limb segments, denoted (F_{thigh1} , F_{thigh2}) on the thigh and (F_{shank1} , F_{shank2}) on the shank. The perpendicular distance between the two forces within each couple is assumed to be identical for both limb segments and equal to d . This yields

$$F_{\text{thigh1}} = F_{\text{thigh2}} = F_{\text{shank1}} = F_{\text{shank2}}. \quad (4)$$

Because each force pair constitutes a pure couple, the net interaction force on each limb segment is zero, and the mechanical effect of the LA configuration is fully characterized by the resulting knee extension torque:

$$\tau_{\text{exo}} = F_{\text{thigh1}} \cdot d. \quad (5)$$

This pure-torque representation is exact within the two-dimensional sagittal-plane model adopted in this study. Unlike the AN and PO configurations, the LA system does not introduce direct compressive or shear force components at the knee joint; any changes in knee contact force arise solely through altered muscle coordination and joint kinematics.

2.1.3 Posterior linkage-driven configuration

In a PO system, a linear actuator located posterior to the knee exerts forces on the posterior aspects of the thigh and shank through attachments (Fig. 1(c)). The thigh and shank interaction forces form an equilibrate pair:

$$F_{\text{thigh}} = F_{\text{shank}}. \quad (6)$$

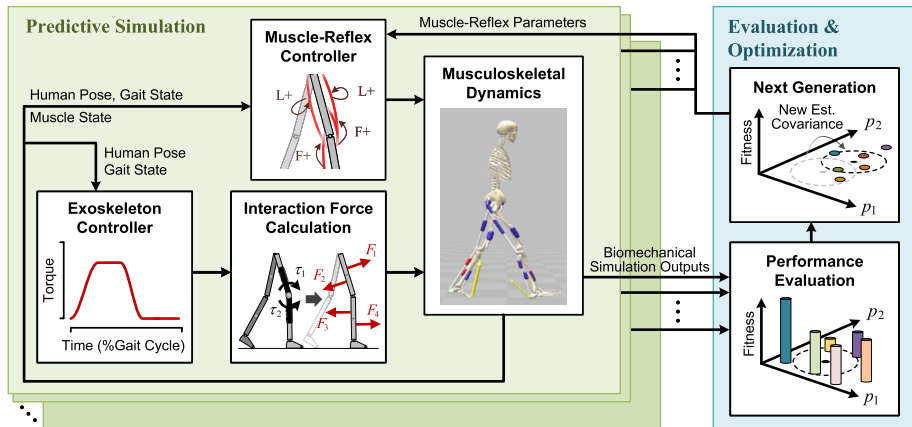


Fig. 2 Predictive simulation framework for the coupled musculoskeletal-exoskeletal system. The framework consists of two interconnected modules. *Predictive Simulation* (green): for each candidate set of muscle-reflex parameters, a forward dynamic simulation is performed, during which the musculoskeletal model (driven by the muscle-reflex controller) and the exoskeleton interaction forces (driven by the exoskeleton controller) jointly determine the resulting biomechanical simulation outputs. *Evaluation & Optimization* (blue): simulation outputs are processed to compute a scalar fitness value using a multi-objective cost function. An optimizer then estimates (Est.) a new population of muscle-reflex parameter sets, which are fed back into the predictive simulation module. This iterative loop continues until convergence toward a dynamically feasible and energetically efficient gait for each exoskeleton configuration and walking scenario

Then the equivalent assistive torque about the knee is

$$\tau_{\text{exo}} = F_{\text{thigh}} \cdot r_{\perp}(\theta_{\text{knee}}), \quad (7)$$

where $r_{\perp}(\theta_{\text{knee}})$ is the perpendicular moment arm of the force relative to the knee joint, and may vary with θ_{knee} due to posterior geometry. A mathematical derivation of $r_{\perp}(\theta_{\text{knee}})$ is given in Appendix A.

2.2 Predictive simulation framework

Figure 2 illustrates the workflow of the predictive simulation framework developed for the coupled musculoskeletal-exoskeletal system in this study. The framework is organized into two major modules: the *Predictive Simulation* module (green region) integrates the muscle-reflex controller, musculoskeletal dynamics, exoskeleton controller, and interaction-force model to execute forward simulation and produces the resulting biomechanical states; the *Evaluation & Optimization* assigns a fitness value to each simulation and updates the distribution of muscle-reflex parameters, producing a new population of parameter sets ($n = 12$) for the next generation. This iterative loop continues until the gait pattern converges to an optimal solution for the given scenario.

2.2.1 Predictive simulation

This module performs forward simulation of the system for each set of muscle-reflex parameters. At each simulation timestep, the muscle-reflex controller receives the current model state, including human gait, ground-contact status, and muscle states, and generates muscle

excitation signals [26]. These excitations drive the musculoskeletal dynamics, where muscle contraction dynamics and skeletal rigid-body equations are integrated to update the whole-body motion.

In parallel, the exoskeleton applies the prescribed knee-extension assistive torque through the controller. This torque is converted into configuration-specific interaction forces through the analytical formulations introduced in Sect. 2.1 and then applied to the human model during the dynamic update. The resulting biomechanical outputs, including joint trajectories, muscle activations, muscle forces, and ground reaction forces, are then forwarded to the *Evaluation & Optimization* module.

The computational framework utilizes the H0918 musculoskeletal model [24], with a total body mass of 74.53 kg and a height of 1.68 m. The skeletal topology constitutes 8 rigid segments, with the head-arms-trunk (HAT) assembly rigidly coupled to the pelvis. The model is configured with 9 generalized degrees of freedom (DOFs) restricted to the sagittal plane: three global DOFs describing pelvic mobility (horizontal and vertical translations, and the anterior-posterior tilt) and six internal DOFs representing the bilateral flexion/extension of the hip, knee, and ankle joints. Actuation is provided by 18 Hill-type muscle-tendon units (9 per leg), encompassing the gluteus maximus, iliopsoas, hamstrings, biceps femoris short head, rectus femoris, vasti, gastrocnemius, soleus, and tibialis anterior. Foot-ground interactions are modeled using two viscoelastic Hunt-Crossley contact spheres located at the heel and toe of each foot. The muscle-reflex controller follows the architecture in [21], which modulates muscle excitations based on kinematic and proprioceptive feedback to yield stable, robust, and biomimetic gait dynamics.

2.2.2 Evaluation and optimization

The evaluation component extracts the relevant biomechanical outputs after each forward dynamic simulation and computes a scalar fitness value (J) through a composite cost function. Following [21], the optimization problem is formulated as

$$\begin{aligned} \min_{\theta} \quad & J = w_1 J_{\text{vel}} + w_2 J_{\text{ang}} + w_3 J_{\text{GRF}} + w_4 J_{\text{CoT}} \\ \text{s.t.} \quad & 0 \leq \sigma(t) \leq 1, \\ & a_{\min} \leq a(t) \leq 1, \\ & \theta_{\min} \leq \theta \leq \theta_{\max}, \end{aligned} \quad (8)$$

where θ denotes the vector of muscle-reflex parameters, $\sigma(t)$ and $a(t)$ represent muscle excitation and activation, respectively, and the minimum activation level is set to $a_{\min} = 0.01$. The state constraints on muscle excitation and activation are inherently enforced by the muscle model and its activation dynamics during forward simulations, while the parameter bounds are enforced during the optimization process, where candidate solutions are sampled within prescribed parameter ranges (see Appendix C for a complete specification of the parameter bounds). These constraints together ensure the physiological plausibility and numerical stability of the multibody simulations throughout the optimization process.

The individual cost terms quantify different aspects of gait performance: J_{vel} quantifies deviations from the prescribed walking velocity; J_{ang} penalizes excessive flexion and extension at the ankle and knee joints; J_{GRF} imposes a threshold penalty activated only when the instantaneous vertical GRF exceeds 1.4 times body weight, serving primarily to prevent unrealistic gait patterns during early optimization iterations; J_{CoT} represents the cost

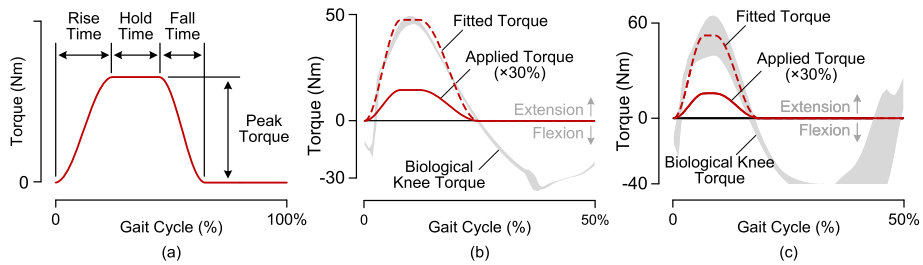


Fig. 3 Parameterization and generation of assistive knee-extension torque over a normalized gait cycle. (a) Parameterized torque profile, characterized by rise time, hold time, fall time, and peak torque. (b) Torque profile for level-ground walking. (c) Torque profile for uphill walking. In (b) and (c), biological knee torque (gray band, mean $\pm$ SD), fitted template (red dashed), and applied assistive torque at 30% amplitude (red solid) are depicted

of transport (CoT), defined as the whole-body metabolic expenditure per unit distance traveled, computed using the metabolic energy model proposed in [27]. Because a common target walking speed is enforced across all conditions, the distance remains effectively constant. Under this condition, CoT is approximately proportional to mass-specific metabolic expenditure and therefore serves as a proxy for whole-body metabolic cost in subsequent analyses.

The weighting coefficients w_1 , w_2 , w_3 , and w_4 are empirically selected and set to 100, 0.1, 10, and 0.1, respectively. These values serve a dual purpose: they partially normalize the cost terms, whose raw magnitudes differ by up to two orders of magnitude, and they encode the intended optimization priority. In particular, the large value of w_1 reflects both the relatively small magnitude of J_{vel} and the high priority of achieving the prescribed walking speed, which is a prerequisite for a physically meaningful predictive gait. Moreover, once the velocity error drops below 0.05 m/s, J_{vel} is set to zero by the optimizer, after which J_{ang} and J_{CoT} become the primary cost terms guiding further parameter refinement.

Based on the fitness values, the optimizer employs the covariance matrix adaptation evolution strategy (CMA-ES) to generate a new population of muscle-reflex parameter sets [28], which are subsequently fed back to the *Predictive Simulation* module for re-evaluation. Through the iterative loop of simulation, evaluation, and parameter update, the system progressively converges toward a dynamically feasible and energetically efficient gait for each exoskeleton configuration and walking scenario. The optimization terminates when the progress in fitness drops below 10^{-5} or when the maximum number of generations (i.e., 10^5) is reached.

2.3 Simulations for different scenarios

Simulations are performed in SCONE (v2.4.0) with the Hyfydy plugin [24]. Two terrains are considered: level-ground and uphill (5%-gradient incline), each simulated for 20 s at a desired walking speed of 1.0 m/s. Under each scenario, predictive simulations are conducted for the unassisted case (UN) and three bilateral exoskeleton-assisted configurations (AN, LA, and PO).

Figure 3(a) illustrates the profile of the assistive knee-extension torque normalized to the gait cycle, parameterized by four variables: rise time, hold time, fall time, and peak torque. The three temporal parameters are expressed as percentages of the gait cycle, defined as the interval between two consecutive heel strikes of the same foot [29]. The rising and

falling phases of the profile are realized using sinusoidal interpolation to guarantee first-derivative continuity. Figure 3(b) and (c) depict the torque-generation procedure for level-ground and uphill walking, respectively. For each terrain, a parameterized template (red dashed line) is fitted to the positive (extension) biological knee torque from the UN condition (gray band, mean $\pm$ SD) during stance. The exoskeleton torque applied to each knee joint of the musculoskeletal model is then prescribed as 30% of the fitted torque (red solid line), consistent with [30]. For each terrain, the same torque profile is used for all three exoskeleton designs to enable controlled cross-configuration comparisons. Because the template was derived solely from the positive stance-phase knee torque and varies smoothly over the gait cycle, the resulting assistance is unidirectional and has modest bandwidth requirements, making it compatible with all three configurations and with common lower-limb exoskeleton actuation systems [31].

To evaluate the sensitivity of the simulation outcomes to key modeling choices, two supplementary analyses are performed for level-ground walking. First, the exoskeleton assistance magnitude is increased by 10% to examine whether the inter-configuration ranking depends on the particular assistance level. Second, an alternative set of optimization weights is adopted in the objective function Eq. (8), where the largest weight is halved and the two smallest weights are doubled (i.e., $w_1 = 50$, $w_2 = 0.2$, $w_3 = 10$, $w_4 = 0.2$) to assess whether the ranking is sensitive to weight selection. In both analyses, all other simulation parameters are held constant.

2.4 Data analysis

For each walking scenario, kinematic and kinetic outcomes, including hip, knee, and ankle joint angles and torques, and ground reaction forces (GRFs), are analyzed across all conditions (UN, AN, LA, and PO). To assess the knee joint loading, the knee extension moment impulse (KEMi) is calculated as the time integral of the knee extension moment over the periods of positive sagittal-plane knee moments [32], i.e.,

$$\text{KEMi} = \frac{1}{m} \int_0^T \tau_{\text{knee}}(t)_+ dt, \quad (9)$$

where $\tau_{\text{knee}}(t)$ denotes the sagittal-plane knee joint moment, T is the gait cycle duration, and $(\cdot)_+$ denotes the positive-part operator defined as $(x)_+ = \max(x, 0)$, and m is the body mass of the model used for normalization.

Muscular responses are evaluated by examining the peak force and muscular effort of the quadriceps, hamstrings, and gastrocnemius, which are selected as representative knee-spanning muscles due to their functional significance during the stance phase. The muscular effort of muscle i , denoted by E_i , is defined as the time integral of its force over one gait cycle [33], i.e.,

$$E_i = \frac{1}{m} \int_0^T F_i(t) dt, \quad (10)$$

where $F_i(t)$ denotes the force generated by muscle i as a function of time t , and T represents the gait cycle duration.

Knee contact forces are compared across scenarios in terms of peak magnitude and timing. The reported knee contact force is defined as the resultant magnitude of the sagittal-plane tibiofemoral joint reaction force, computed using an approach equivalent to the *Joint Reaction Analysis* in OpenSim [34]. In this method, the distal body segment is modeled as

an independent rigid body with known kinematics in a global reference frame. The joint reaction forces and moments at the body origin $\mathbf{R}_o$ are obtained via Newton–Euler force balance:

$$\mathbf{R}_o = \begin{bmatrix} \boldsymbol{\tau}_o \\ \mathbf{F}_o \end{bmatrix} = \mathbf{M}_i(\mathbf{q}) \mathbf{a}_i + \mathbf{F}_{\text{constraint}} - \left(\sum \mathbf{F}_{\text{external}} + \sum \mathbf{F}_{\text{muscles}} + \mathbf{R}_{i+1} \right), \quad (11)$$

where $\mathbf{M}_i(\mathbf{q})$ is the 3×3 spatial mass matrix of the body segment i , $\mathbf{a}_i$ is its three-dimensional acceleration vector (one angular and two linear components in the sagittal plane), $\mathbf{F}_{\text{constraint}}$ represents any constraint forces, $\mathbf{F}_{\text{external}}$ and $\mathbf{F}_{\text{muscles}}$ are the applied external loads and musculotendon forces, respectively, $\mathbf{R}_{i+1}$ is the previously computed reaction load at the distal joint, $\boldsymbol{\tau}_o$ is the moment component, and $\mathbf{F}_o$ is the translational force component. The reported knee contact force is the Euclidean norm of the translational component $\|\mathbf{F}_o\|$, which includes contributions from all musculotendon forces, exoskeleton interaction forces, ground reaction forces, gravitational loads, and segment inertial effects. This formulation has been validated against in vivo tibiofemoral contact force measurements from instrumented knee implants, showing good agreement in both magnitude and temporal profiles during gait [35–37].

CoT is extracted from the optimization objective J_{CoT} in Eq. (8) to assess whole-body metabolic cost under each condition. Configuration-specific human-exoskeleton interaction forces are also extracted from the simulation outputs: for each configuration, the peak strap force is reported, defined as the interaction force magnitude at the thigh or shank attachment site ($F_{\text{thigh}} = F_{\text{shank}}$ for AN and PO; any one of the four equal forces for LA); for the AN configuration, the knee redirection force F_{knee} is additionally reported. All time-dependent data are normalized to the gait cycle and reported as a percentage of the gait cycle (0–100%GC). All force- and moment-related quantities are normalized to the model's body mass.

3 Results

3.1 Level-ground walking

For all four level-ground walking conditions, the CMA-ES optimizer converged well within the maximum budget of 10^5 generations, requiring 505 (UN), 846 (AN), 1272 (LA), and 786 (PO) generations, respectively. These values correspond to at most 1.3% of the allowed budget, indicating that the optimization problem is well-conditioned across configurations for this terrain. The converged gait solutions are analyzed in the following subsections.

3.1.1 Kinematics and kinetics

In Figs. 4–7, each condition is distinguished by both color and linestyle (UN: gray dashed; AN: orange solid; LA: yellow long-dashed; PO: purple double-dashed). Alternating gray bands divide the gait cycle into seven sub-phases [29], with abbreviated labels provided in the phase bar at the upper left of the figure (LR = Loading Response, MSt = Mid-Stance, TSt = Terminal Stance, PSw = Pre-Swing, ISw = Initial Swing, MSw = Mid-Swing, TSw = Terminal Swing).

Across all assisted conditions, joint kinematics exhibit a consistent temporal advancement relative to the UN condition (Fig. 4(a)). For the knee, peak flexion during Loading

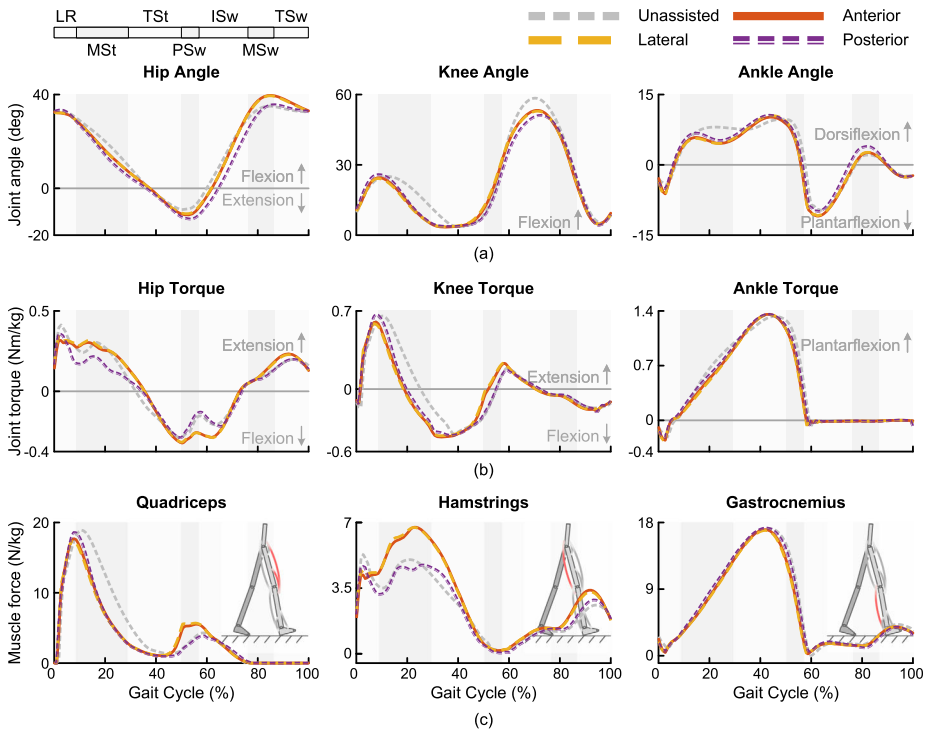


Fig. 4 Kinematic, kinetic, and muscular outputs during level-ground walking for unassisted (UN, gray dashed), anterior (AN, orange solid), lateral (LA, yellow long-dashed), and posterior (PO, purple double-dashed) conditions. Data are normalized to the gait cycle. (a) Hip, knee, and ankle joint angles. (b) Corresponding joint torques, normalized to body mass. (c) Quadriceps, hamstrings, and gastrocnemius muscle forces, normalized to body mass. Alternating light and dark gray bands demarcate seven gait sub-phases, labeled in the phase bar at the upper left: LR = Loading Response, MSt = Mid-Stance, TSt = Terminal Stance, PSw = Pre-Swing, ISw = Initial Swing, MSw = Mid-Swing, TSw = Terminal Swing [29]. This color/linestyle convention and phase-band scheme are used consistently in Figs. 4–7 and Fig. 9–12

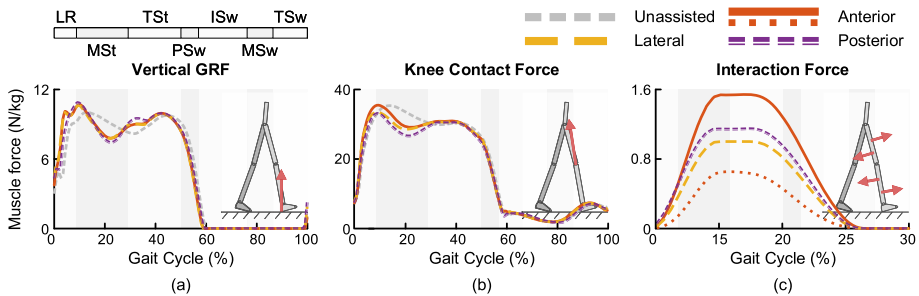


Fig. 5 (a) Vertical GRFs, (b) knee contact forces, and (c) human-exoskeleton interaction forces during level-ground walking for unassisted, anterior, lateral, and posterior conditions. Data are time-normalized to the gait cycle and magnitude-normalized to body mass. Linestyles, colors, and gait sub-phase bands follow the convention of Fig. 4. The orange dotted line in (c) represents the knee redirection force specific to the anterior condition

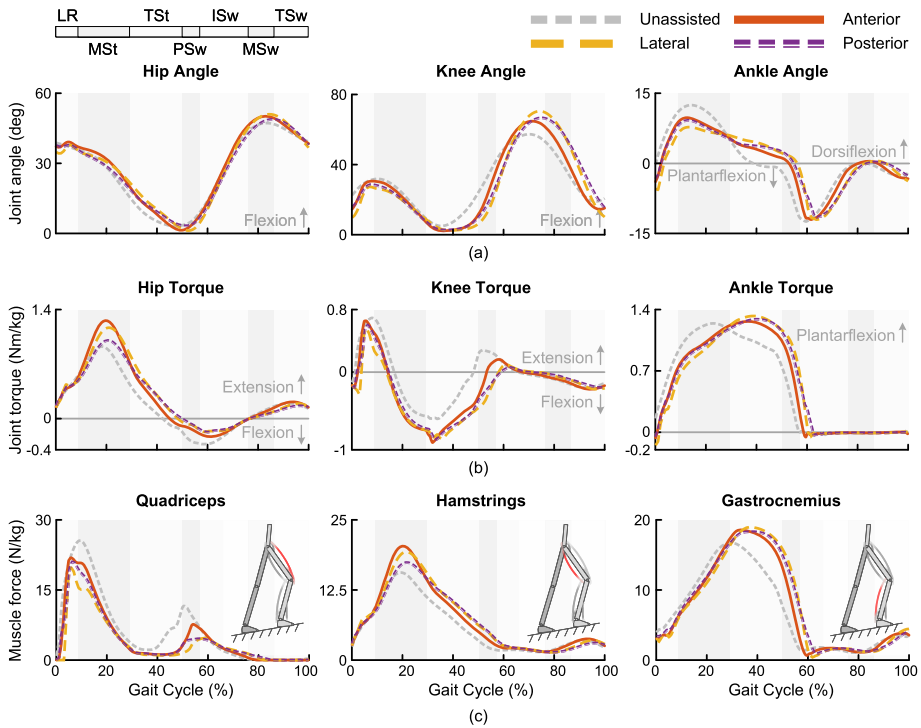


Fig. 6 Kinematic, kinetic, and muscular outputs during uphill walking for unassisted, anterior, lateral, and posterior conditions. Data are normalized to the gait cycle. (a) Hip, knee, and ankle joint angles. (b) Corresponding joint torques, normalized to body mass. (c) Quadriceps, hamstrings, and gastrocnemius muscle forces, normalized to body mass. Linestyles, colors, and gait sub-phase bands follow the convention of Fig. 4

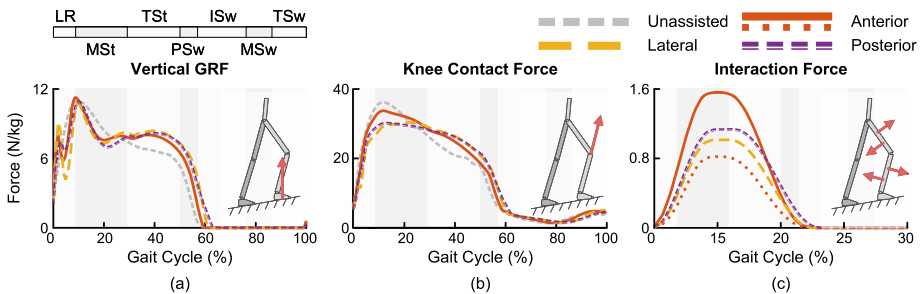


Fig. 7 (a) Vertical GRFs, (b) knee contact forces, and (c) human-exoskeleton interaction forces during uphill walking for unassisted, anterior, lateral, and posterior conditions. Data are time-normalized to the gait cycle and magnitude-normalized to body mass. Linestyles, colors, and gait sub-phase bands follow the convention of Fig. 4. The orange dotted line in (c) represents the knee redirection force specific to the anterior condition

Response occurs earlier under assistance (UN 12.3%GC; AN / LA / PO 8.6%GC). The transition toward the state of terminal extension (i.e., knee flexion angle ≤ 5 deg) during Mid-Stance also shifts earlier (UN 36.4%GC; AN 29.2%GC; LA 29.4%GC; PO 29.8%GC). Hip motion shows a similar advancement, with the neutral position reached earlier under assis-

tance (UN 33.0%GC; AN 31.5%GC; LA 31.6%GC; PO 29.3%GC). For the ankle, both the dorsiflexion peak and subsequent valley reach earlier with assistance (peak: UN 22.2%GC, AN 14.8%GC, LA 14.7%GC, PO 15.0%GC; valley: UN 34.0%GC, AN 24.0%GC, LA 23.2%GC, PO 24.3%GC).

Joint torques are shown in Fig. 4(b). KEMi is 0.102 Nm·s/kg in the UN condition and decreases to 0.0706, 0.0676, and 0.0825 Nm·s/kg in AN, LA, and PO, corresponding to reductions of 30.8%, 33.7%, and 19.1%, respectively. Peak knee extension torque also occurs earlier across all assisted configurations (UN 24.8%GC; AN 18.5%GC; LA 18.1%GC; PO 19.3%GC). In terms of magnitude, PO exhibits a slightly higher peak knee extension torque (0.669 Nm/kg) relative to UN (0.654 Nm/kg), whereas AN (0.598 Nm/kg) and LA (0.584 Nm/kg) show reduced peaks. The timing of the extension-to-flexion torque transition follows the same advancement (UN 24.8%GC; AN 18.5%GC; LA 18.1%GC; PO 19.3%GC). At the hip, PO produces reduced extension torque compared with UN, whereas AN and LA show higher flexion torque from Terminal Stance to Initial Swing. For the ankle, plantarflexion torque peaks slightly earlier under assistance (UN 45.7%GC; AN 43.2%GC; LA 43.6%GC; PO 43.3%GC).

Vertical GRF (Fig. 5(a)) shows consistent temporal and magnitude changes under assistance. During Loading Response, all three assisted conditions present a higher GRF peak (UN 10.0 N/kg; AN 10.7 N/kg; LA 10.6 N/kg; PO 10.9 N/kg) and an earlier peak time (UN 12.5%GC; AN 10.2%GC; LA 9.5%GC; PO 9.4%GC). The timing of the second GRF peak remains largely unchanged, though its magnitude shows minimal variation.

3.1.2 Muscle forces and cost of transport

Figure 4(c) shows muscle forces of the quadriceps, hamstrings, and gastrocnemius. The muscles spanning the knee exhibit distinct responses to assistance across scenarios. For the quadriceps, peak force decreases under AN and LA (UN 18.7 N/kg; AN 17.7 N/kg; LA 17.4 N/kg), whereas PO shows a smaller reduction (18.6 N/kg). Muscular effort decreases in all assisted conditions (UN 5.47 N·s/kg; AN 4.73 N·s/kg; LA 4.72 N·s/kg; PO 4.57 N·s/kg). Peak quadriceps force occurs earlier under assistance (UN 10.7%GC; AN 7.8%GC; LA 7.0%GC; PO 7.8%GC).

For the hamstrings, muscular effort increases under the AN and LA conditions relative to UN (2.80 N·s/kg), reaching 3.59 N·s/kg and 3.58 N·s/kg, respectively, whereas PO produces a lower increase (2.87 N·s/kg). These changes are accompanied by elevated force levels during Mid-Stance in AN and LA, while PO remains comparable to UN throughout most of the gait cycle. The gastrocnemius force profile exhibits minimal changes, with both peak force and muscular effort remaining close to UN across all assisted configurations.

CoT reflects the net metabolic consequence of the muscle consequence of the muscle force redistribution described above. All three assisted configurations achieve lower CoT than UN: PO achieves the lowest CoT (4.37 J/(m·kg)), representing a 7.02% reduction relative to UN (4.70 J/(m·kg)); AN and LA produce more modest reductions of 1.49% (4.63 J/(m·kg)) and 0.851% (4.66 J/(m·kg)), respectively.

3.1.3 Knee contact forces and interaction forces

Knee contact force peaks shift earlier under all assistance conditions compared with UN (Fig. 5(b)). A corresponding shift in the knee joint kinematics is also observed in Fig. 4(a). Peak magnitudes differ across conditions, with AN showing the highest value (35.5 N/kg), followed by UN (35.4 N/kg), LA (33.5 N/kg), and PO (33.1 N/kg). Apart from the peak

region, knee contact force profiles remain similar across conditions, with only minor variations in shape and timing.

Configuration-specific interaction forces during assistance are shown in Fig. 5(c). The AN configuration produces the highest peak strap force of 1.53 N/kg (114 N), followed by PO at 1.15 N/kg (85.7 N) and LA at 1.00 N/kg (74.6 N). In addition, AN introduces a posterior-directed knee redirection force with a peak of 0.655 N/kg (48.8 N), arising from cable redirection at the knee pulley.

3.1.4 Sensitivity and robustness

Two supplementary analyses were performed for level-ground walking (see Sect. 2.3) to examine the robustness of the inter-configuration rankings to key variations in assistance magnitude and optimization weights. The rankings are largely preserved under both perturbations. The few deviations are limited to reversals between AN and LA in metrics for which the two configurations yield nearly identical values, and are therefore attributed to negligible numerical differences rather than substantive changes in biomechanical behavior. Consequently, the comparative conclusions remain unchanged under the tested variations. Detailed results are provided in Appendix D.

3.2 Uphill walking

For uphill walking, convergence was likewise achieved rapidly, at 1227 (UN), 1113 (AN), 745 (LA), and 517 (PO) generations, all remaining below 1.3% of the maximum budget of 10^5 generations. The efficient convergence across both terrains suggests that the cost function landscape is smooth and that the reported solutions are not sensitive to the generation budget. The biomechanical and metabolic outcomes for each condition are presented below.

3.2.1 Kinematics and kinetics

Joint kinematics during uphill walking exhibit systematic temporal changes under all assisted conditions relative to UN (Fig. 6(a)). Prominent differences appear at the knee: although kinematics remain similar to UN in Mid-Stance, all assisted conditions exhibit a prolonged near-neutral position (≤ 5 deg) in Terminal Stance, as indicated by a delayed transition back to flexion (UN 35.2%GC; AN 45.7%GC; LA 48.7%GC; PO 45.8%GC). This extended state of terminal extension contrasts with level-ground walking, where neutral crossing occurs earlier with assistance. Hip kinematics remain largely similar among conditions with only small timing differences. At the ankle, the plantarflexion peak in Initial Swing is delayed under all assisted conditions (UN 59.6%GC; AN 61.6%GC; LA 63.8%GC; PO 64.0%GC).

Joint torque profiles (Fig. 6(b)) demonstrate reduced knee extension loading under assistance. KEMi decreases from 0.0755 Nm·s/kg in UN to 0.0515, 0.0297, and 0.0401 Nm·s/kg in AN, LA, and PO, corresponding to reductions of 31.8%, 60.7%, and 46.9%, respectively. In contrast to level-ground walking scenarios, where PO produces the smallest KEMi reduction, here AN yields the least KEMi reduction during uphill walking, with LA and PO showing more pronounced decreases. A similar reordering is observed in peak knee extension torque, with uphill peak values highest in UN (0.698 Nm/kg), followed by AN (0.660 Nm/kg), PO (0.605 Nm/kg), and LA (0.559 Nm/kg). This contrasts with the level-ground results, where PO exhibits the highest peak, followed by UN, AN, and LA. Additionally, the timing of the extension-to-flexion transition is advanced under assisted conditions (UN

16.8%GC; AN 14.3%GC; LA 13.7%GC; PO 14.6%GC), consistent with the timing shifts observed in level-ground walking scenarios. In Terminal Stance, hip, knee, and ankle torques are elevated under assistance relative to UN.

Vertical GRF during uphill walking (Fig. 7(a)) exhibits a systematic delay in the second GRF rise and unloading under all assisted conditions relative to UN. This delay contrasts with level-ground walking, where stance-phase timing remains similar across conditions.

3.2.2 Muscle forces and cost of transport

Muscle force profiles for uphill walking are presented in Fig. 6(c). For the quadriceps, the highest peak force occurs in the UN condition (25.6 N/kg), followed by AN (21.9 N/kg), PO (21.0 N/kg), and LA (19.8 N/kg). This ordering differs from the level-ground walking results, where PO produces a peak force comparable to UN. Correspondingly, quadriceps muscular effort is reduced under all assisted conditions, decreasing from 6.77 N·s/kg in UN to 5.04 N·s/kg, 3.96 N·s/kg, and 4.34 N·s/kg in AN, LA, and PO, respectively.

The hamstrings exhibit elevated muscular effort across the assisted conditions relative to UN (6.29 N·s/kg), with AN and LA producing the largest increases (7.65 N·s/kg and 7.56 N·s/kg), and PO showing a more moderate increase (6.96 N·s/kg). For the gastrocnemius, a delayed peak occurs under assistance, accompanied by higher muscular effort compared with UN (UN 7.44 N·s/kg; AN 8.14 N·s/kg; 8.23 N·s/kg; PO 8.03 N·s/kg).

CoT reduced under all assisted conditions relative to UN (7.26 J/(m·kg)): LA achieves the lower CoT (6.85 J/(m·kg), a 5.65% reduction), followed by PO (6.99 J/(m·kg), 3.72%) and AN (7.03 J/(m·kg), 3.17%). Notably, the CoT ranking across configurations is consistent with that of quadriceps muscular effort, despite the elevated hamstrings and gastrocnemius demands observed under assistance.

3.2.3 Knee contact forces and interaction forces

Knee contact force characteristics during uphill walking are illustrated in Fig. 7(b). Peak values decrease under all assisted configurations relative to UN (UN 54.2 N/kg; AN 50.6 N/kg; LA 45.5 N/kg; PO 45.3 N/kg). This outcome differs from level-ground walking, where AN produces a slightly higher peak than UN, although the ranking among the assisted conditions remains consistent across terrains (i.e., AN produces the largest peak, followed by LA and PO).

Interaction forces during are shown in Fig. 7(c). The configuration ranking is consistent with level-ground walking: AN produces the highest peak strap force of 1.57 N/kg (117 N), followed by PO at 1.14 N/kg (84.8 N) and LA at 1.02 N/kg (75.8 N). The AN knee redirection force increases to 0.822 N/kg (61.3 N), compared with 0.655 N/kg (48.8 N) during level-ground walking. This is consistent with the larger knee flexion angles observed (Fig. 6(a)), which increase the cable redirection angle according to Eq. (2). Numerical values of peak knee extension torque, KEMi, peak quadriceps force, muscular effort for the three muscles, peak knee contact force, peak interaction forces, and CoT for all scenarios are summarized in Appendix B.

4 Discussion

4.1 Summary of key findings

This study systematically evaluated the biomechanical effects of three knee exoskeleton assistance configurations during level-ground and uphill walking using a unified predictive

musculoskeletal simulation framework. All conditions converged within 517–1272 generations, well below the maximum budget (10^5), indicating that the optimization problem is well-conditioned across all configurations and terrains. To the best of our knowledge, this is the first study to directly compare multiple exoskeleton configurations across walking conditions within an identical musculoskeletal model. Across both terrains, all assisted conditions consistently reduced the KEMi and CoT. Additionally, assisted walking was characterized by systematic alterations in the timing and coordination of lower limb biomechanics, including advanced or prolonged joint postures, altered muscle force patterns, and shifts in joint torques during the stance phase of gait. Notably, assistance altered the profiles of knee joint torques, reflecting a redistribution of extension and flexion demands rather than a simple attenuation of peak knee loading. Collectively, these findings demonstrate that knee exoskeleton assistance, even under the idealized device and joint modeling assumptions adopted, induces global, task-dependent adaptations in gait mechanics that extend beyond local knee joint unloading.

Despite these common trends, the magnitude and assistance effects were strongly dependent on both exoskeleton configuration and walking condition. During level-ground walking, the posterior configuration exhibited a smaller reduction in knee extension effort relative to the anterior and lateral configurations, whereas this ordering was altered during uphill walking, where the anterior configuration showed the least pronounced reduction. Similar configuration- and terrain-dependent reordering were observed in quadriceps muscular effort and knee contact force, suggesting that assistance effects characterized during level-ground walking do not directly generalize to inclined locomotion.

4.2 Mechanical interpretation of gait adaptations

The observed gait adaptations under knee exoskeleton assistance reflect coordinated neuromuscular responses to externally applied forces, rather than simple mechanical offloading at the knee. Although the exoskeletons considered in this study were designed to provide solely knee extension assistance, their effects propagated beyond the knee to adjacent joints and muscle groups. These cross-joint adaptations arise from the intrinsic mechanical coupling of the lower-limb musculoskeletal system, primarily mediated by multi-articular muscles. Muscles such as rectus femoris, hamstrings, and gastrocnemius, which were examined in this study, span both the knee and adjacent joints, thereby mechanically linking knee assistance to hip and ankle mechanics [29]. Consequently, reductions in knee extensor demand were accompanied by systematic changes in hip and ankle kinematics, joint torques, and muscle force profiles across both level-ground and uphill walking conditions.

Another key manifestation of this coordination is a redistribution of knee extension and flexion demands during stance. Across assisted conditions, the terminal knee extension state occurred earlier than in the unassisted condition, which was also reported in prior experimental studies of knee exoskeleton assistance [7, 38]. From a functional perspective, shifting knee loading from active extension toward controlled flexion may offer mechanical advantages during stance. In particular, a portion of the flexion torque can be supported by passive joint structures and geometric locking mechanisms, rather than requiring continuous activation of knee flexor muscles [39]. Such a strategy reduces reliance on active muscular force production while maintaining joint stability during Loading Response and Mid-Stance. Consistent with this view, the cost of transport is reduced under all assisted conditions on both terrains, and its ranking across configurations closely follows that of quadriceps muscular effort rather than that of the compensating muscle groups. This indicates that, despite the compensatory demands placed on the hamstrings and gastrocnemius, the net metabolic outcome remains governed primarily by the degree of quadriceps offloading. This redistribution

suggests that exoskeleton assistance reshapes not only the magnitude of knee loading, but also the mechanical pathways through which knee torques are generated.

Notably, the uphill walking simulations revealed a concurrent reduction in quadriceps effort together with increased hamstrings and gastrocnemius effort under knee assistance. This muscle-level redistribution closely aligns with experimental observations reported by [7], where similar compensation patterns were observed during assisted uphill walking with knee exosuits. The agreement between the present predictive simulations and prior experimental findings provides additional support for the physiological plausibility of the predicted neuromuscular adaptations. Although the quantitative magnitude of these effects may be modulated by device mass and inertia properties not captured in the present model, the consistency with experimental observations suggests that the simulations capture the essential neuromuscular coordination mechanisms underlying assisted gait.

More importantly, the concurrent changes in joint kinematics and ground reaction forces indicate that altered muscle coordination and timing directly influenced whole-body dynamics. This highlights a limitation of conventional inverse-dynamics approaches based on experimentally prescribed kinematics, which are inherently unable to capture assistance-induced adaptations in gait strategy and load redistribution [21]. In contrast, the muscle-reflex-based predictive simulation framework employed in this study allows such adaptations to emerge naturally, providing mechanical insight into how knee exoskeleton assistance interacts with human locomotor control.

4.3 Effects of terrain and exoskeleton configuration

Beyond the general mechanisms underlying assistance-induced gait adaptations, the magnitude and distribution of biomechanical responses are strongly dependent on exoskeleton configuration and walking terrain. During level-ground walking, the anterior and lateral configurations achieved greater reductions in knee extension effort than the posterior configuration, whereas the posterior configuration outperformed the anterior configuration during uphill walking. This reversal indicates that configuration-specific effects substantially depend on how assistive forces interact with joint posture and task demands, rather than on nominal assistive torque alone.

These differences can be understood by considering the distinct force application mechanisms of the three configurations. The lateral configuration primarily provides a direct knee extension torque and exhibits relatively stable performance across joint postures. In contrast, the anterior and posterior configurations rely on force redirection and are therefore more sensitive to knee flexion angle. The anterior cable-driven configuration generates knee extension assistance mainly through a posteriorly directed knee force resulting from cable routing. This mechanism is mechanically advantageous when knee flexion angles are small, but its effective moment arm decreases as knee flexion increases. Conversely, the posterior linkage-driven configuration produces assistance through posteriorly applied force couples that not only generate knee extension torque, but also introduce a vertical support component, with mechanical effectiveness of extension assistance increasing at larger knee flexion angles. This vertical support component also helped reduce the hip extension torque demand during stance compared with other configurations.

These complementary posture-dependent characteristics explain the observed terrain-specific reordering. During level-ground walking, the knee remains close to terminal extension in the stance phase, favoring the anterior configuration while limiting the effective leverage of posterior assistance. During uphill walking, larger knee flexion angles in Mid-Stance naturally enhance the mechanical advantage of posterior assistance while reducing

the effectiveness of anteriorly routed cables. As a result, posterior assistance provides greater unloading of hip-and-knee extension effort in uphill walking, whereas anterior assistance becomes comparatively less effective. The lateral configuration, being less sensitive to knee posture, shows more consistent performance across terrains.

Knee contact force differences across configurations can be interpreted by considering the combined contributions of ground reaction forces, muscle forces, and externally exerted forces. The anterior configuration applies a downward force to the thigh and an upward force to the shank, introducing a compressive component across the knee and partially counteracting muscle-related unloading. In contrast, the posterior configuration generates a supportive force couple to the lower limb of the wearer, thereby directly reducing knee joint compression [40]. The lateral configuration does not provide a direct external unloading effect, but achieves comparable reductions in knee contact force primarily through greater decreases in muscle-generated joint loading. From a clinical perspective, configurations that reduce knee loading while limiting contact forces may offer distinct advantages for populations with compromised joint health, such as older adults or individuals with knee osteoarthritis [41].

Taken together, these findings emphasize that the biomechanical effectiveness of knee exoskeleton assistance is inherently task- and configuration-dependent. From a design and selection perspective, these results underscore the importance of jointly considering walking conditions, force application geometry, and their downstream effects on joint loading and muscle coordination. Configurations that perform favorably under level-ground walking may not retain the same advantages during inclined locomotion, and vice versa. Predictive simulation frameworks that allow gait adaptations to emerge across tasks thereby offer a valuable tool for identifying configuration-specific strengths and limitations prior to experimental deployment.

4.4 Implications of human-exoskeleton interaction forces

Across both terrains, the AN configuration consistently exhibits the highest peak strap forces (114–117 N). This arises from its small cable offset relative to the limb axis, which requires a large cable tension to generate the prescribed assistive torque. In contrast, the LA configuration requires the lowest interaction forces, as its force-couple mechanism distributes torque generation over the brace separation distance d , which is typically larger than the moment arms in the AN and PO configurations.

Beyond differences in force magnitude, the manner in which interaction forces are transmitted to the biological limb is critical for practical implementation. The estimated force levels are not only technically achievable but also fall safety within the physiological comfort limits characterized in recent literature: these values remain well below the total pulling force limits reported for the lower limbs (867 ± 296 N for the thigh and 702 ± 220 N for the shank) [42], and stay within the range of localized compression forces (230–340 N) known to be tolerable with standardized padding [43].

However, satisfying these global magnitude constraints does not inherently guarantee user comfort, as interaction experience is heavily dictated by the anatomical region of loading. While the force levels are theoretically safe, the AN configuration concentrates the redirection force at the anterior knee region, where soft tissue coverage is limited and sensitivity to pressure is high. In contrast, the LA configuration avoids net translational loading on limb segments, resulting in a more distributed and potentially less intrusive interaction pattern. These findings suggest that while all evaluated configurations maintain realistic and ergonomic force magnitudes, the LA interface offers a superior loading strategy by optimizing the spatial distribution of pressure across the human-exoskeleton interface.

Notably, the configuration associated with the highest interaction forces (AN) does not yield the most favorable biomechanical outcomes across most reported metrics. In contrast, the LA and PO configurations achieve comparable or greater reductions in KEMi, knee contact force, and CoT while requiring lower interface forces. This observation indicates that effective assistance is not solely governed by the magnitude of the applied forces, but also by their orientation and transmission relative to the underlying joint mechanics. Consequently, optimizing configuration and force direction may be as important as increasing actuation capacity in the design of exoskeleton systems.

4.5 Implications of the idealized knee representation

The biological knee was modeled in this study as a single-degree-of-freedom hinge joint with a fixed center of rotation. This represents a best-case assumption for human-exoskeleton kinematic compatibility, because it eliminates the poly-centric motion and migrating instantaneous center of rotation (ICR) that characterize the anatomical knee. Under this idealization, the exoskeleton and biological joint axes remain perfectly aligned throughout the gait cycle, and no parasitic interaction forces arise from kinematic mismatch.

The practical implications of this assumption differ across configurations. In the AN and PO configurations, assistance is transmitted through elements that do not introduce an explicit hinge joint aligned with the biological knee. Consequently, these architectures do not impose a fixed mechanical axis at the knee and are not subject to joint misalignment in the same manner as lateral designs. In contrast, most physical implementations of the LA configuration incorporate a mechanical knee joint intended to align with the biological knee axis, and are therefore inherently sensitive to alignment errors between the device and anatomical joint axes [16]. Even small discrepancies can generate parasitic interaction forces, increased friction, and unintended constraints on natural joint motion, potentially reducing effective torque transmission [44]. Such parasitic loads would also propagate to the knee joint contact force, affecting the predicted contact mechanics. Incorporating a more anatomically realistic knee representation, e.g., a single-degree-of-freedom mechanism with a moving ICR as has been adopted in several biomechanical simulations of knee joint mechanics [45, 46], could therefore reveal additional parasitic loading in the LA configuration that is absent under the present idealized conditions.

These considerations suggest that the biomechanical advantages predicted for the LA configuration may represent an upper bound on achievable performance. Specifically, parasitic interaction forces arising from joint misalignment would reduce effective torque transmission, thereby attenuating the predicted reductions in KEMi; necessitate additional muscular effort to maintain the intended gait pattern, increasing metabolic cost; and alter knee contact force through the superposition of unintended loading components. A more detailed knee model could diminish these advantages and potentially alter the relative ranking among configurations. Accordingly, both biomechanical effectiveness and human-device kinematic compatibility should be considered jointly when interpreting the present results and when selecting exoskeleton architectures for practical deployment.

4.6 Limitations and future work

Several limitations of this study should be acknowledged. A notable limitation of this study is that the mass, inertia, and transmission losses of the exoskeleton hardware were not modeled. This simplification affects all three configurations but is particularly consequential for cross-configuration comparisons, as the three configurations differ substantially

in how device mass is distributed relative to the lower-limb segments. For instance, rigid lateral-actuation designs concentrate considerable mass at or around the knee joint, whereas cable-driven devices typically locate the actuator mass on the waist or thigh, resulting in a more proximal mass distribution. Consequently, the metabolic and biomechanical benefits reported here represent an idealized upper bound and may be attenuated in practice, especially for configurations with greater distal mass. Specifically, the added limb inertia would require greater muscular effort during the swing phase, increasing metabolic cost and potentially altering the optimized gait pattern; moreover, given that LA configuration introduces a greater mass concentration at the knee joint compared to the AN and PO configurations, its metabolic benefit is expected to be disproportionately attenuated, which may alter the relative performance ranking among the three configurations. Incorporating realistic, configuration-specific mass and inertia properties, potentially through co-simulation with detailed multibody exoskeleton models, remains an important direction for future work. Future iterations should also incorporate more detailed human-exoskeleton interaction models, including soft tissue deformation, device inertia, and interface slippage, to better reflect real-world mechanics.

Another limitation is the use of a two-dimensional musculoskeletal model in which the biological knee is represented as an ideal hinge joint. This formulation precludes the representation of medio-lateral dynamics, including frontal-plane balance perturbations that may arise during exoskeleton wear, as well as varus-valgus loading at the knee. Consequently, the knee contact force reported in this study is the resultant of only the vertical and anterior-posterior joint reaction force components; the medio-lateral contribution is absent, and the reported values therefore underestimate the true three-dimensional joint reaction force magnitude. In addition, since exoskeleton structures alter the mass distribution about the limb and may impose laterally directed interaction forces, frontal-plane balance demands during assisted gait could differ across configurations in ways that the present sagittal-plane analysis cannot capture. The idealized hinge-joint assumption further neglects secondary knee kinematics such as tibial rotation and condylar translation, which may affect joint contact mechanics and are relevant for assessing potential misalignment between the biological and the exoskeleton joint axes. As discussed in Sect. 4.5, these model simplifications are particularly consequential for the generalizability of the predicted configuration ranking to practical scenarios. Extending the current framework to a three-dimensional musculoskeletal model with a more anatomically detailed knee representation constitutes an important direction for future work.

Moreover, the current simulations considered only a mild 5% incline. This gradient was selected for several reasons: it represents a commonly encountered slope in daily community ambulation (e.g., accessible ramps, sidewalks), and the reflex-based locomotion controllers used in predictive neuromuscular simulations have been primarily validated for level and mildly inclined walking, with their robustness at steeper gradients not yet well established. Importantly, prior experimental studies indicate that joint-level biomechanical demands evolve approximately monotonically with increasing gradient [47–49], suggesting that the configuration-dependent trends identified at 5% are likely to be preserved or amplified at steeper inclines. Nevertheless, future work should evaluate a broader range of gradients to verify this expectation and to test the boundaries of the predictive framework.

Finally, the AN configuration was modeled by assuming that the interaction forces act parallel to their respective limb segments at an equal offset r from both the thigh and shank axes. In a physical implementation, the cable protrusion may differ between the two segments and the forces may not remain parallel to the limbs, altering the effective moment arm. More broadly, refining muscle-reflex strategies and conducting targeted experimental

validation will be essential to translate these configuration-specific insights into practical design and control guidelines for knee exoskeletons. Such validation should compare predicted gait kinematics, ground reaction forces, and muscle activation timing against experimental measurements obtained from motion capture, force plates, and surface electromyography, respectively, and further compare predicted metabolic cost against respirometry data, for subjects walking with and without each exoskeleton configuration. These comparisons would help to verify that the muscle-reflex control structure and the optimization criterion employed produce physiologically plausible neuromuscular coordination patterns under exoskeleton assistance.

5 Conclusion

This study presented a systematic comparison of three representative knee exoskeleton configurations using a unified predictive musculoskeletal simulation framework across level-ground and uphill walking. By applying analytically derived, configuration-specific interaction forces within an identical musculoskeletal model, this framework enabled a direct comparison of the biomechanical consequences associated with different actuation pathways under varying terrains. The results demonstrate that, under the idealized massless device assumption adopted in this study, while all configurations effectively reduced knee extension effort and cost of transport, the magnitude and nature of biomechanical benefits were strongly configuration- and terrain-dependent. Distinct reordering of effectiveness was observed across knee extension moment impulse, quadriceps effort, and knee contact force between level-ground and uphill walking scenarios, indicating that assistance performance cannot be inferred from a single metric or walking condition. These differences arise from the interaction between assistive force geometry, joint posture, and whole-body mechanics, rather than nominal assistive torque alone. Collectively, these findings underscore the importance of evaluating knee exoskeletons as task-dependent human–device systems. The proposed predictive simulation framework provides a mechanical and scalable tool for identifying configuration-specific strengths and limitations prior to experimental deployment, offering actionable insight for the design of knee exoskeletons tailored to specific locomotor tasks.

Appendix A: Derivation of assistive forces in posterior linkage-driven configuration

As shown in Fig. 8(a), the posterior linkage-driven (PO) exoskeleton configuration generates assistance through a pair of opposing forces applied to the posterior aspects of the thigh and shank, denoted as F_{thigh} and F_{shank} . For force analysis, this interaction was represented using an equivalent geometric model (Fig. 8(b)). The knee joint center is denoted by point O , with force application points at A on the thigh and B on the shank. Three angles are defined to characterize posture dependence: the knee flexion angle θ_{knee} , the angle between F_{thigh} and the thigh segment θ_{thigh} , and the angle between F_{shank} and the shank segment θ_{shank} . $r_{\perp}$ represents the perpendicular moment arm of the forces relative to the knee joint O . Additionally, $AC \perp OC$ and $BD \perp OD$.

By the Pythagorean theorem, we obtain the following relationship:

$$|OA| = \sqrt{|OC|^2 + |AC|^2}, \quad |OB| = \sqrt{|OD|^2 + |BD|^2}. \quad (12)$$

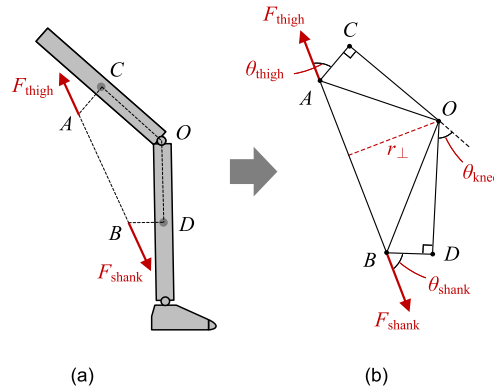


Fig. 8 Illustration of the knee exoskeleton of PO configuration and its equivalent geometric representation used for force analysis. (a) Conceptual illustration of the posterior configuration interacting with the human lower limb. The exoskeleton applies a pair of opposing forces on the posterior aspects of the thigh and shank, denoted as F_{thigh} and F_{shank} . (b) Simplified geometric model used to describe the force transmission mechanism and its dependence on joint posture. Key points denote the knee joint center (O), force application locations (A and B), and geometric constraints (C and D). The knee joint angle θ_{knee} , the angle between F_{thigh} and the thigh segment θ_{thigh} , and the angle between F_{shank} and the shank segment θ_{shank} , are defined. The perpendicular moment arm of the forces relative to O is denoted as $r_{\perp}$

The angle $\angle AOB$ is given by

$$\begin{aligned}\angle AOB &= \pi - \theta_{knee} - \angle AOC - \angle BOD \\ &= \pi - \theta_{knee} - \arctan(|AC|/|OC|) \\ &\quad - \arctan(|BD|/|OD|).\end{aligned}\quad (13)$$

Applying the law of cosines to triangle $\triangle OAB$, we get

$$|AB| = \sqrt{|OA|^2 + |OB|^2 - 2|OA||OB|\cos\angle AOB}, \quad (14)$$

$$\angle OAB = \arccos\left(\frac{|OA|^2 + |AB|^2 - |OB|^2}{2|OA||OB|}\right). \quad (15)$$

Therefore, the perpendicular moment arm $r_{\perp}$ is given by

$$r_{\perp} = |OA|\sin\angle OAB. \quad (16)$$

The points C and D are assumed to coincide with the centers of mass of the thigh and shank segments, respectively. Accordingly, $|OC|$ and $|OD|$ represent the distances from the knee joint center (point O) to the segment centers of mass, which were obtained from the musculoskeletal model (the H0918 model [24]): $|OC| = 0.26$ m and $|OD| = 0.1867$ m. The distances between the force application points and the segment centers, $|AC|$ and $|BD|$, were both set to 0.13 m, representing a practically feasible placement of the interaction forces on the thigh and shank, consistent with typical values reported in the literature [12].

Appendix B: Summary of biomechanical and metabolic metrics

This subsection summarizes key quantitative outcome metrics used throughout the study to facilitate cross-condition comparison for level-ground walking (Table 1) and uphill walking

Table 1 Summary of Biomechanical and Metabolic Metrics During Level-Ground Walking

Item	Unit	UN	AN	LA	PO
Peak knee extension torque	Nm/kg	0.654	0.598	0.584	0.669
KEMi	Nm·s/kg	0.102	0.0706	0.0676	0.0825
Peak quadriceps force	N/kg	18.7	17.7	17.4	18.6
Quadriceps muscular effort	N·s/kg	5.47	4.73	4.72	4.57
Hamstrings muscular effort	N·s/kg	2.80	3.59	3.58	2.87
Gastrocnemius muscular effort	N·s/kg	7.67	7.51	7.38	7.93
Peak knee contact force	N/kg	35.4	35.5	33.5	33.1
Peak strap force	N/kg (N)	—	1.53 (114)	1.00 (74.6)	1.15 (85.7)
Peak knee redirection force	N/kg (N)	—	0.655 (48.8)	—	—
CoT	J/(m·kg)	4.70	4.63	4.66	4.37

Note: UN = unassisted walking; AN = anterior cable-driven exoskeleton configuration; LA = lateral parallel-actuated exoskeleton configuration; PO = posterior linkage-driven exoskeleton configuration; KEMi = knee extension moment impulse; CoT = cost of transport.

Table 2 Summary of Biomechanical and Metabolic Metrics During Uphill Walking

Item	Unit	UN	AN	LA	PO
Peak knee extension torque	Nm/kg	0.698	0.660	0.559	0.605
KEMi	Nm·s/kg	0.0755	0.0515	0.0297	0.0401
Peak quadriceps force	N/kg	25.6	21.9	19.8	21.0
Quadriceps muscular effort	N·s/kg	6.77	5.04	3.96	4.34
Hamstrings muscular effort	N·s/kg	6.29	7.65	7.56	6.96
Gastrocnemius muscular effort	N·s/kg	7.44	8.14	8.22	8.03
Peak knee contact force	N/kg	54.2	50.6	45.5	45.3
Peak strap force	N/kg (N)	—	1.57 (117)	1.02 (75.8)	1.14 (84.8)
Peak knee redirection force	N/kg (N)	—	0.822 (61.3)	—	—
CoT	J/(m·kg)	7.26	7.03	6.85	6.99

Note: UN = unassisted walking; AN = anterior cable-driven exoskeleton configuration; LA = lateral parallel-actuated exoskeleton configuration; PO = posterior linkage-driven exoskeleton configuration; KEMi = knee extension moment impulse; CoT = cost of transport.

(Table 2) under all assistance configurations. All values are reported with three significant figures. Note that peak muscle force is reported only for the quadriceps, as it directly relates to knee extension loading and was explicitly analyzed in Sect. 3. For other muscles, muscular effort is reported as a more representative measure of cumulative loading. Peak interaction forces (strap forces and knee redirection forces) are reported in both normalized and absolute units.

Appendix C: Muscle-reflex parameter bounds

This subsection specifies the bounds of muscle-reflex controller parameter during the CMA-ES optimization process, following the conventions of SCONE Tutorial 4a – Gait [24]. The

Table 3 Lower and Upper Bounds of Muscle-Reflex Controller Parameters

Parameter	Lower Bound	Upper Bound	Parameter	Lower Bound	Upper Bound
Activated Throughout Gait Cycle					
tib_ant.KL	0	10	gastroc.L0	0.5	2
tib_ant.L0	0.5	2	gastroc.KF	-10	10
tib_ant.KF	-10	10	tib_ant-soleus.KL	-10	0
soleus.KL	0	10	tib_ant-soleus.KF	-2	0
soleus.L0	0.5	2	tib_ant-gastroc.KL	-10	0
soleus.KF	-10	10	tib_ant-gastroc.KF	-2	0
gastroc.KL	0	10	soleus-tib_ant.KF	-2	0
Activated from Loading Response to Terminal Stance					
vasti.KL	0	4	rect_fem.L0	0	2
vasti.L0	0	2	hamstrings.KL	0	4
vasti.KF	0	4	hamstrings.L0	0	2
iliopsoas.KL	0	4	bifemsh.KF	0	4
iliopsoas.L0	0	2	glut_max.KL	0	4
rect_fem.KL	0	4	glut_max.L0	0	2
gastroc-tib_ant.KF	-2	0	glut_max-pelvis_tilt.KP	-10	10
hamstrings-pelvis_tilt.KP	-10	10	hamstrings-pelvis_tilt.KV	0	10
rect_fem-pelvis_tilt.KP	-10	10	rect_fem-pelvis_tilt.KV	-10	10
Activated from Pre-Swing to Terminal Swing					
iliopsoas.KL	0	10	hamstrings.L0	0	2
iliopsoas.L0	0	2	bifemsh.KF	0	4
rect_fem.KL	0	10	glut_max.KL	-10	10
rect_fem.L0	0	2	glut_max.L0	0	2
iliopsoas-pelvis_tilt.KP	0	10	iliopsoas-pelvis_tilt.KV	0	10
iliopsoas-hamstrings.KL	-10	10	iliopsoas-hamstrings.L0	0	2
hamstrings.KL	-10	10			

corresponding results are summarized in Table 3. Each parameter name combines a muscle identifier and a gait type. The muscle identifier is formatted as *target_muscle* for single-muscle reflexes or *target-source* for cross-muscle reflexes or sensory-driven reflexes. Gain types include KL (length feedback gain), KF (force feedback gain), KP (postural feedback gain), L0 (reference length), and KV (velocity feedback gain). Further details can be found in the SCONE documentation (<https://scone.software>).

Appendix D: Sensitivity and robustness analyses

To evaluate whether the inter-configuration rankings reported in the main text are robust to key modeling choices, two additional sets of simulations are performed for level-ground walking: (i) a 10% increase in exoskeleton assistance magnitude, and (ii) an alternative optimization weight set. In each analysis, only the parameters under examination are modified, whereas all other simulation settings are kept identical to those adopted in the main text.

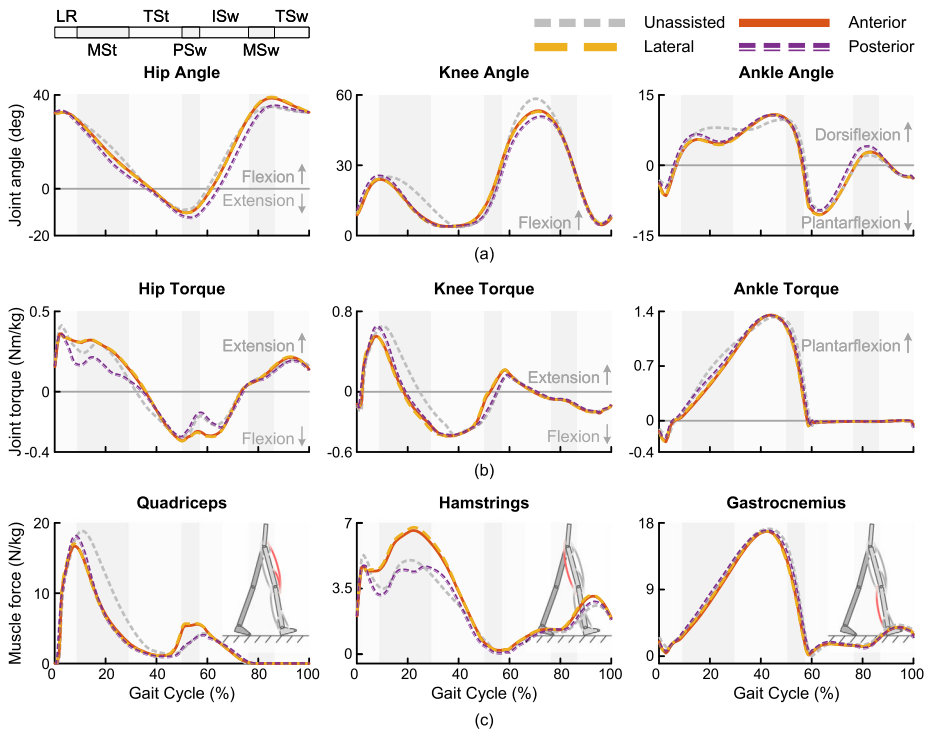


Fig. 9 Kinematic, kinetic, and muscular outputs under altered assistance magnitude during level-ground walking for unassisted, anterior, lateral, and posterior conditions. Data are normalized to the gait cycle. (a) Hip, knee, and ankle joint angles. (b) Corresponding joint torques, normalized to body mass. (c) Quadriceps, hamstrings, and gastrocnemius muscle forces, normalized to body mass. Linestyles, colors, and gait sub-phase bands follow the convention of Fig. 4

D.1 Effect of altered assistance magnitude

Figures 9–10 present the biomechanical results for the unassisted condition and the three assisted configurations under the altered-assistance condition, and Table 4 summarizes the key quantitative metrics. Compared with the baseline results reported in Table 1, the inter-configuration rankings are preserved for peak knee extension torque, KEMi, peak knee contact force, peak strap force, and CoT. For peak quadriceps force and muscular efforts, the rankings of AN and LA are reversed relative to those in the baseline analysis. However, these reversals are associated with numerically negligible inter-configuration differences: in both the baseline and altered-assistance analysis, the difference in peak quadriceps force between AN and LA is within 0.3 N/kg, and the differences in muscular efforts are within 0.13 N·s/kg. This interpretation is further supported by the waveform-level comparison between Fig. 4(c) and Fig. 9(c), in which the muscle force trajectories of AN and LA are nearly superimposed throughout the gait cycle. Accordingly, these ranking reversals are best interpreted as arising from near identical values rather than from a meaningful change in biomechanical response, and they do not alter the substantive conclusions.

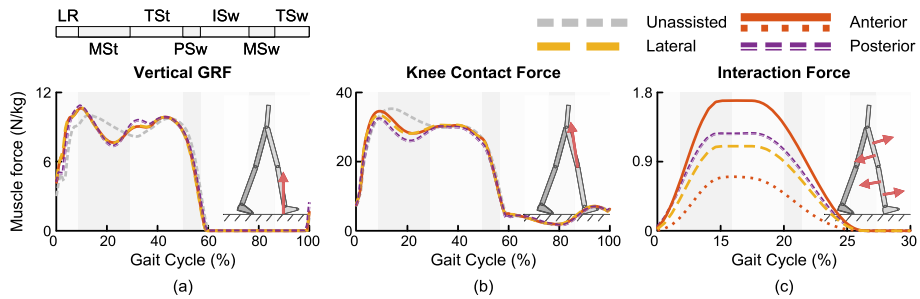


Fig. 10 (a) Vertical GRFs, (b) knee contact forces, and (c) human-exoskeleton interaction forces under altered assistance magnitude during level-ground walking for unassisted, anterior, lateral, and posterior conditions. Data are time-normalized to the gait cycle and magnitude-normalized to body mass. Linestyles, colors, and gait sub-phase bands follow the convention of Fig. 4. The orange dotted line in (c) represents the knee redirection force specific to the anterior condition

Table 4 Summary of Biomechanical and Metabolic Metrics Under Altered Assistance During Level-Ground Walking

Item	Unit	UN	AN	LA	PO
Peak knee extension torque	Nm/kg	0.654	0.554	0.564	0.651
KEMi	Nm·s/kg	0.102	0.0645	0.0643	0.0789
Peak quadriceps force	N/kg	18.7	16.8	17.1	18.1
Quadriceps muscular effort	N·s/kg	5.47	4.56	4.60	4.49
Hamstrings muscular effort	N·s/kg	2.80	3.49	3.57	2.84
Gastrocnemius muscular effort	N·s/kg	7.67	7.15	7.24	7.81
Peak knee contact force	N/kg	35.4	34.6	33.5	32.5
Peak strap force	N/kg (N)	—	1.69 (126)	1.10 (82.1)	1.27 (94.6)
Peak knee redirection force	N/kg (N)	—	0.704 (52.4)	—	—
CoT	J/(m·kg)	4.70	4.57	4.62	4.36

Note: UN = unassisted walking; AN = anterior cable-driven exoskeleton configuration; LA = lateral parallel-actuated exoskeleton configuration; PO = posterior linkage-driven exoskeleton configuration; KEMi = knee extension moment impulse; CoT = cost of transport.

D.2 Effect of altered optimization weights

Figures 11–12 show the biomechanical results for the unassisted condition and the three assisted configurations using the alternative optimization weight set. The waveform patterns and the inter-configuration separation remain qualitatively similar to the baseline results. Key quantitative metrics are reported in Table 5. The inter-configuration rankings are preserved across all metrics except hamstrings and gastrocnemius muscular efforts, where AN and LA are reversed. As in the altered-assistance analysis (Sect. D.1), these reversals occur because the two configurations yield nearly identical values and the observed differences fall within the range attributable to the modest shift in optimization landscape. These reversals are therefore likewise interpreted as numerically negligible and not indicative of substantive change in predicted biomechanical behavior.

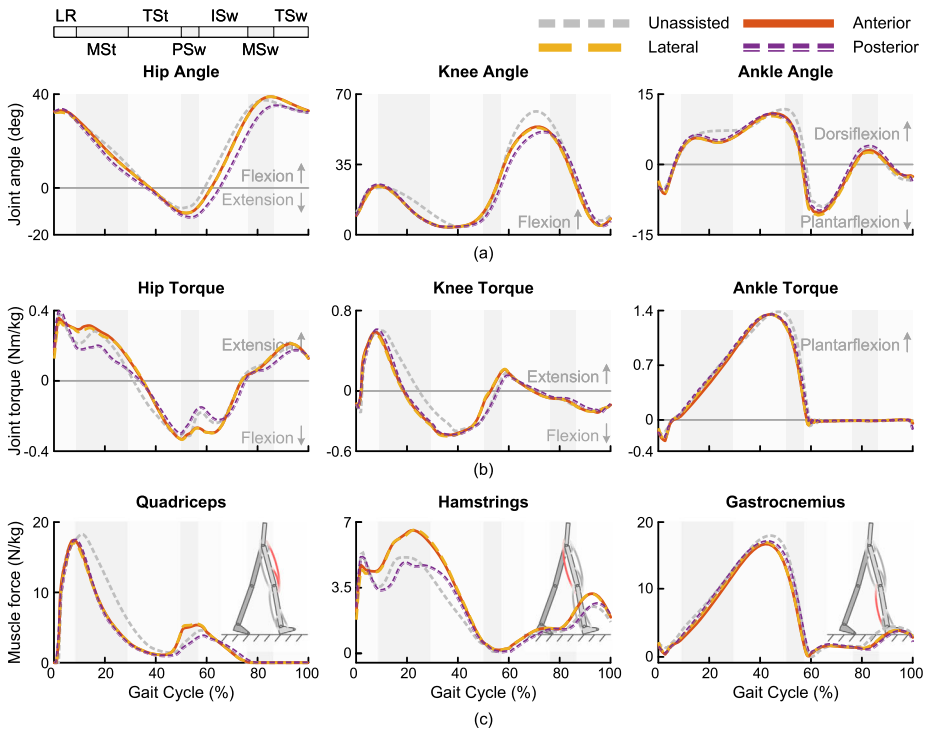


Fig. 11 Kinematic, kinetic, and muscular outputs under altered optimization weights during level-ground walking for unassisted, anterior, lateral, and posterior conditions. Data are normalized to the gait cycle. (a) Hip, knee, and ankle joint angles. (b) Corresponding joint torques, normalized to body mass. (c) Quadriceps, hamstrings, and gastrocnemius muscle forces, normalized to body mass. Linestyles, colors, and gait sub-phase bands follow the convention of Fig. 4

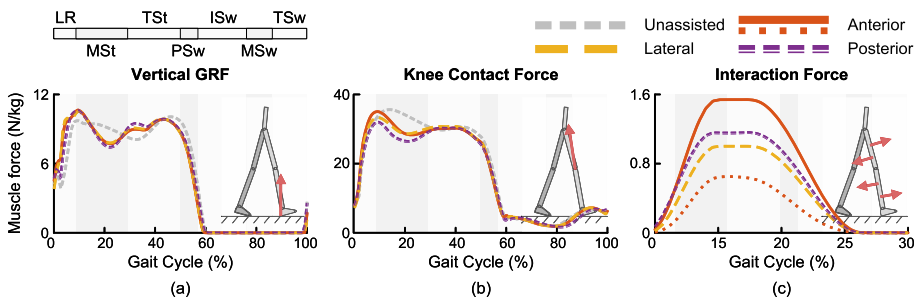


Fig. 12 (a) Vertical GRFs, (b) knee contact forces, and (c) human-exoskeleton interaction forces under altered optimization weights during level-ground walking for unassisted, anterior, lateral, and posterior conditions. Data are time-normalized to the gait cycle and magnitude-normalized to body mass. Linestyles, colors, and gait sub-phase bands follow the convention of Fig. 4. The orange dotted line in (c) represents the knee redirection force specific to the anterior condition

Table 5 Summary of Biomechanical and Metabolic Metrics Under Altered Optimization Weights During Level-Ground Walking

Item	Unit	UN	AN	LA	PO
Peak knee extension torque	Nm/kg	0.600	0.588	0.566	0.610
KEMi	Nm·s/kg	0.0967	0.0703	0.0662	0.0745
Peak quadriceps force	N/kg	18.3	17.5	17.1	17.5
Quadriceps muscular effort	N·s/kg	5.56	4.69	4.66	4.38
Hamstrings muscular effort	N·s/kg	2.73	3.50	3.52	2.84
Gastrocnemius muscular effort	N·s/kg	8.06	7.17	7.42	7.70
Peak knee contact force	N/kg	35.6	35.1	33.4	32.0
Peak strap force	N/kg (N)	—	1.54 (115)	1.00 (74.6)	1.16 (86.4)
Peak knee redirection force	N/kg (N)	—	0.650 (48.5)	—	—
CoT	J/(m·kg)	4.89	4.60	4.64	4.33

Note: UN = unassisted walking; AN = anterior cable-driven exoskeleton configuration; LA = lateral parallel-actuated exoskeleton configuration; PO = posterior linkage-driven exoskeleton configuration; KEMi = knee extension moment impulse; CoT = cost of transport.

D.3 Summary

Taken together, the two analyses show that the inter-configuration rankings reported in the main text are largely preserved. The few ranking reversals that do occur are limited to AN and LA with minimal differences and do not affect the overall conclusions. Accordingly, the relative advantages of each configuration in reducing knee extension efforts, knee contact force, quadriceps loading and metabolic cost, as well as the associated trade-offs in interaction forces, remain unchanged under the tested variations in assistance magnitude and optimization weight set. These results indicate that they are governed primarily by the underlying biomechanical mechanisms rather than by specific parameter tunings within the simulation framework.

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Declarations

Competing interests The authors declare no competing interests.

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